## ASSIGNMENT-2

# PCB DESIGN PROJECT REPORT

Push Button LED and Buzzer Circuit (Emergency Alarm Unit)

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| **Assignment** | Assignment-2 |
| **Project Title** | Push Button LED and Buzzer Circuit using KiCad |
| **EDA Software** | KiCad |

**Problem statement:** A company wants to install emergency alarm units in its office. When a panic button is pressed, a buzzer and warning LED should activate to alert nearby personnel. As a PCB Design Engineer, the task is to design this module using KiCad.

This report documents the circuit schematic, the footprint assignments, the PCB layout (2D) and 3D views, the bill of materials, the project objectives and the conclusion.

**Status note:** The PCB layout screenshot still shows thin cyan connection guides (ratsnest lines), which usually indicate connections that are not yet routed with copper tracks. The layout is therefore presented as the placement view. Gerber and drill files were not supplied.

# 1. Schematic

The circuit is powered from a 5 V input at the two-pin screw terminal (J1). The push button (SW1) controls the path to the LED and buzzer branch. The schematic shows an LED (D1), two 220 Ω resistors (R1 and R2), and a buzzer (BZ1). Verify the buzzer's rated voltage/current and the final wiring before powering the assembled board.

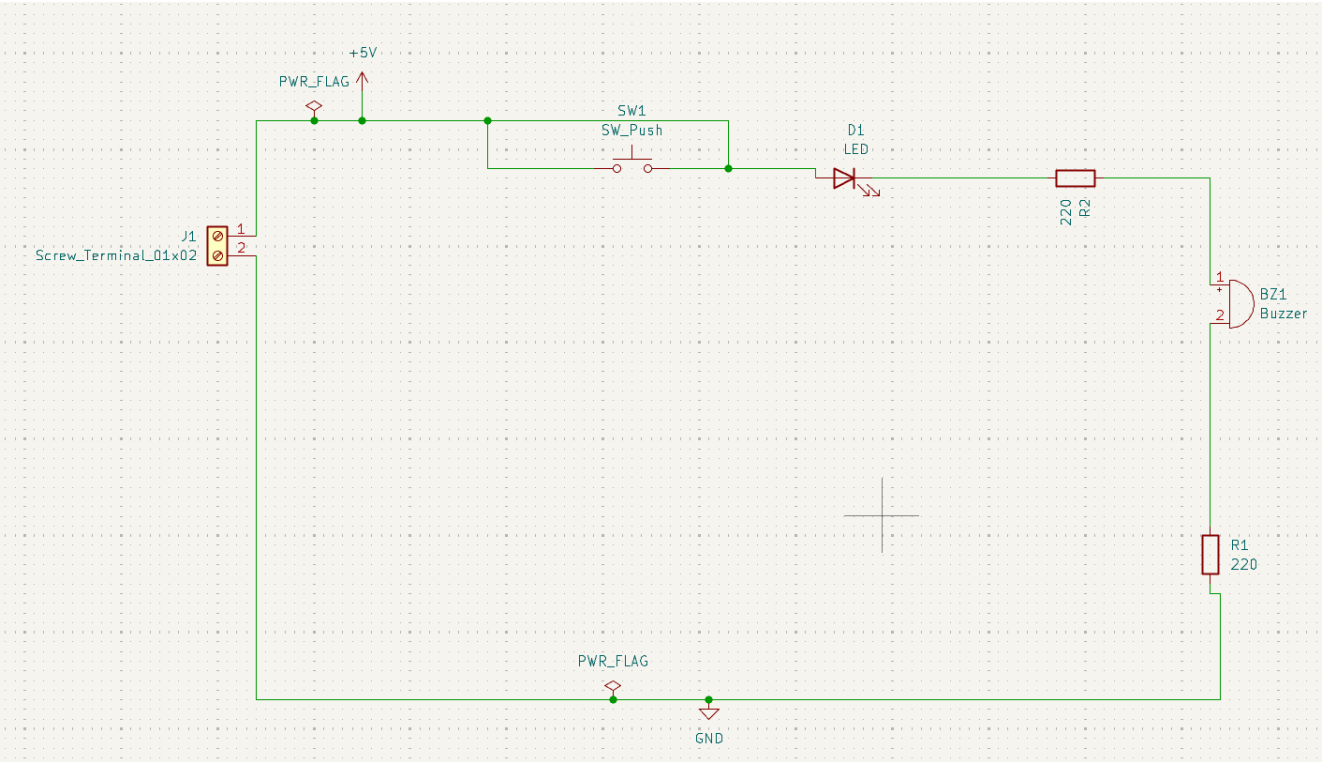



Figure 1. Schematic of the push button LED and buzzer circuit.

## 2. PCB Footprints

Each schematic symbol was assigned a through-hole footprint using the Symbol : Footprint Assignments tool in KiCad, as shown below.

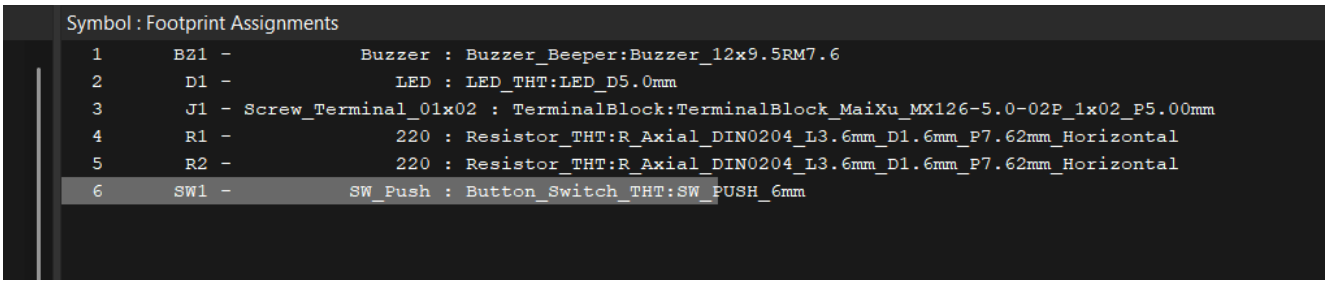



Figure 2. Symbol : Footprint Assignments.

| Ref | Component | Footprint |
|---|---|---|
| BZ1 | Buzzer | Buzzer_Beeper:Buzzer_12x9.5RM7.6 |
| D1 | LED | LED_THT:LED_D5.0mm |
| J1 | Screw_Terminal_01x02 | TerminalBlock:TerminalBlock_MaiXu_MX126-5.0-02P_1x02_P5.00mm |
| R1, R2 | 220 Ω resistor | Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm_P7.62mm_Horizontal |
| SW1 | SW_Push | Button_Switch_THT:SW_PUSH_6mm |

Table 1. Footprint summary.

## 3. 2D PCB View

The PCB layout below shows the placed footprints for the screw terminal, buzzer, push button, LED and the two resistors, with reference designators and net names. The cyan airwires show the connections that still need copper-track routing.

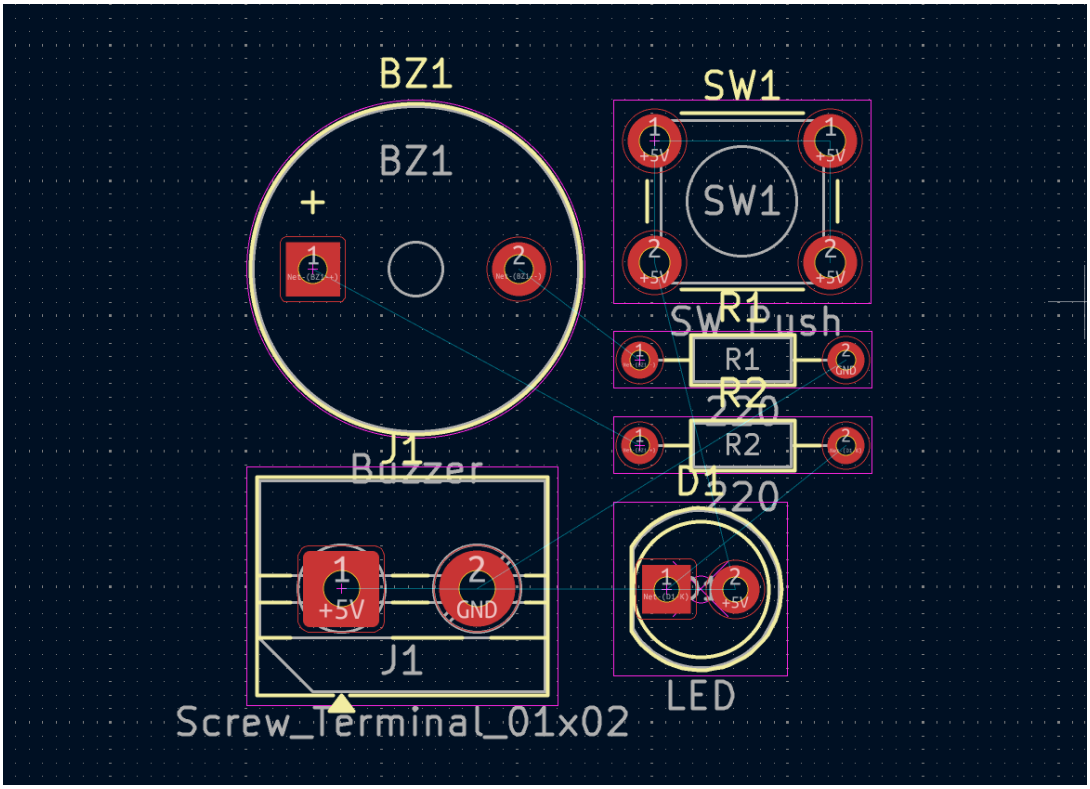



Figure 3. PCB layout (2D) in KiCad PCB Editor.

## 4. 3D PCB View

The 3D viewer shows the board with the component models: buzzer, push button SW1, resistors R1 and R2, LED D1 and screw terminal J1.

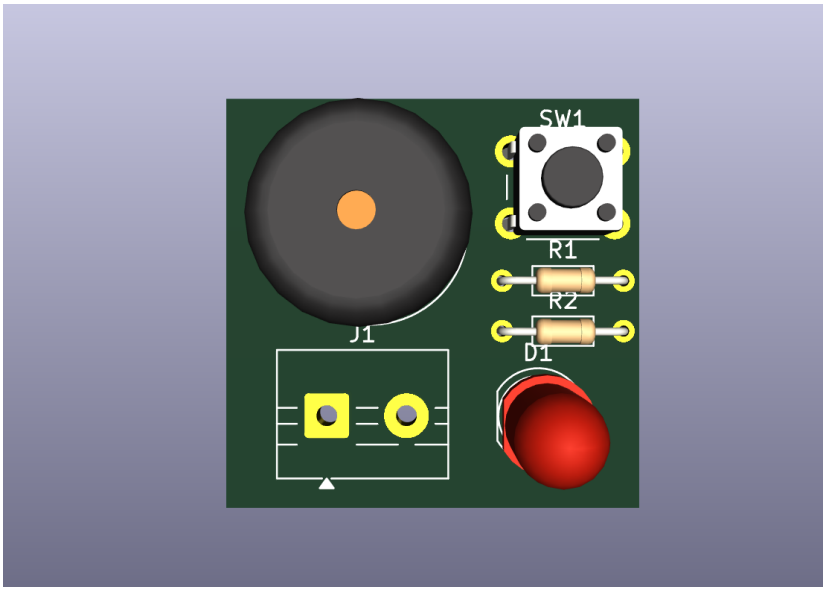



Figure 4. 3D view of the PCB in KiCad 3D Viewer.

# 5. Bill of Materials (BOM)

| Reference | Component | Value / Type | Qty. | Notes |
|---|---|---|---|---|
| J1 | 2-pin screw terminal | Screw_Terminal_01x02 | 1 | 5 V input / GND |
| SW1 | Push button switch | SW_Push | 1 | Momentary switch |
| D1 | LED | LED | 1 | Colour not specified |
| R1 | Resistor | 220 Ω | 1 | Power rating not specified |
| R2 | Resistor | 220 Ω | 1 | Power rating not specified |
| BZ1 | Buzzer | Buzzer | 1 | Exact model not specified |

BOM is based on the schematic labels. Confirm the exact LED colour, resistor power rating and buzzer part number before purchasing.

# 6. Objectives of the Project

- To design a compact emergency alarm unit in which pressing a panic button activates a buzzer and a warning LED to alert nearby personnel.
- To draw the circuit schematic in KiCad using a 5 V supply through a two-pin screw terminal (J1) and a push button (SW1) as the trigger.
- To limit the current through the LED and buzzer branch using 220 Ω resistors (R1 and R2).
- To assign suitable through-hole footprints to all components and place them on a single PCB layout.
- To review the design through the 2D PCB layout and the 3D viewer before manufacturing.
- To prepare the design for fabrication by completing track routing, running the Design Rules Checker (DRC) and exporting Gerber and drill files.

# 7. Conclusion

The push button LED and buzzer circuit for the emergency alarm unit was captured as a schematic in KiCad, all components were assigned footprints, and the board was placed and checked in the 2D layout and 3D viewer. The design meets the functional requirement that a press of the panic button powers the buzzer and the LED together from a 5 V input.

To make the design ready for manufacture, the remaining copper tracks should be routed, the DRC run, and the Gerber and Excellon drill files exported from the final PCB.